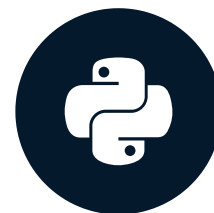


# Numeric data types

DATA TYPES IN PYTHON



**Jason Myers**  
Instructor

# Built in numeric types

## Integer

- Whole numbers
- Large values

```
int(123456789123456789)
```

```
123456789123456789
```

## Float

- Fractional amounts (approximation)
- Scientific notation

```
float(123456789123456789)
```

```
1.2345678912345678e+17
```

# Decimals

- Exact precision
- Currency operations

```
from decimal import Decimal  
Decimal('123456789123456789')
```

```
Decimal('123456789123456789')
```

# Printing floats

```
print(0.00001)
```

```
1e-05
```

```
print(f"{0.00001:f}")
```

```
0.000010
```

# Printing floats

```
print(f"{0.00000001:f}")
```

```
0.000000
```

```
print(f"{0.00000001:.7f}")
```

```
0.0000001
```

# Python division types

```
4/2
```

```
2.0
```

```
4//2
```

```
2
```

```
7//3
```

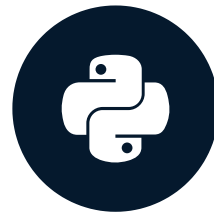
```
2
```

# Let's practice!

DATA TYPES IN PYTHON

# Booleans - the logical data type

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# Booleans as a data type

- True
- False

Notice the capitalization as this can trip you up when switching between Python and other languages.

```
out_of_cookies = True
if out_of_cookies:
    print("Run to the store NOW!")
```

```
Run to the store NOW!
```

# Truthy and Falsey

- Truthy values are ones that will return true
- Falsey values will evaluate to false

```
apples=2  
if apples:  
    print("We have apples.")
```

```
"We have apples."
```

```
apples=0  
if apple:  
    print('We have apples.')
```

# Truthy and Falsey

## Truthy

- `1`
- `"Cookies"`
- `["Cake", "Pie"]`
- `{"key": "value"}`

## Falsey

- `0`
- `""`
- `[]`
- `{}`
- `None`

# Operators - a boolean evaluation context

```
cookie_qty == 3
```

- `==` equal to
- `!=` not equal to
- `<` less than
- `<=` less than or equal to
- `>` greater than
- `>=` greater than or equal to

# Floats are approximately an issue

```
x = 0.1 + 1.1  
x == 1.2
```

```
False
```

```
print(x)
```

```
1.20000000000000000002
```

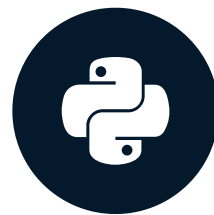
Be careful with equality comparisons of floats!

# Let's practice!

DATA TYPES IN PYTHON

# Sets (unordered data with optimized logic operations)

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# Set

- Unique
- Unordered
- Mutable
- Python's implementation of Set Theory from Mathematics

# Creating sets

- Sets are created from a list

```
cookies_eaten_today = ['chocolate chip', 'peanut butter',  
    ...: 'chocolate chip', 'oatmeal cream', 'chocolate chip']  
types_of_cookies_eaten = set(cookies_eaten_today)  
print(types_of_cookies_eaten)
```

```
set(['chocolate chip', 'oatmeal cream', 'peanut butter'])
```

# Modifying sets

- `.add()` adds single elements

```
types_of_cookies_eaten.add('biscotti')
```

```
types_of_cookies_eaten.add('chocolate chip')
```

```
print(types_of_cookies_eaten)
```

```
set(['chocolate chip', 'oatmeal cream', 'peanut butter', 'biscotti'])
```

# Updating sets

- `.update()` merges in another set or list

```
cookies_hugo_ate = ['chocolate chip', 'anzac']  
  
types_of_cookies_eaten.update(cookies_hugo_ate)  
  
print(types_of_cookies_eaten)
```

```
set(['chocolate chip', 'anzac', 'oatmeal cream', 'peanut butter'])
```

# Removing data from sets

- `.discard()` safely removes an element from the set by value
- `.pop()` removes and returns an arbitrary element from the set (KeyError when empty)

```
types_of_cookies_eaten.discard('biscotti')  
print(types_of_cookies_eaten)
```

```
set(['chocolate chip', 'anzac', 'oatmeal cream', 'peanut butter'])
```

```
types_of_cookies_eaten.pop()  
types_of_cookies_eaten.pop()
```

```
'chocolate chip'  
'anzac'
```

# Set operations - similarities

- `.union()` set method returns a set of all the names (`or`)
- `.intersection()` method identifies overlapping data (`and`)

```
cookies_jason_ate = set(['chocolate chip', 'oatmeal cream',  
    'peanut butter'])  
cookies_hugo_ate = set(['chocolate chip', 'anzac'])  
cookies_jason_ate.union(cookies_hugo_ate)
```

```
set(['chocolate chip', 'anzac', 'oatmeal cream', 'peanut butter'])
```

```
cookies_jason_ate.intersection(cookies_hugo_ate)
```

```
set(['chocolate chip'])
```

# Set operations - differences

- `.difference()` method identifies data present in the set on which the method was used that is not in the arguments (`-`)
- Target is important!

```
cookies_jason_ate.difference(cookies_hugo_ate)
```

```
set(['oatmeal cream', 'peanut butter'])
```

```
cookies_hugo_ate.difference(cookies_jason_ate)
```

```
set(['anzac'])
```

# Let's practice!

DATA TYPES IN PYTHON